JanGyan — Learn Without Limits

Category 3: Road Safety & Public Safety

Module 5 - Two-Wheeler & Four-Wheeler Safety

📋 Module Overview

This handbook is your complete, beginner-to-advanced guide to **Two-Wheeler & Four-Wheeler Safety** — the equipment, restraints, documents, and maintenance habits that protect you every time you ride or drive. Whether you are a new rider, a parent buckling in a child, or a car owner managing paperwork and servicing, this handbook is written so that you can read it once and never be unsure about what keeps you legally and physically safe.

By the end of this module, you will not need to search anywhere else to understand vehicle and occupant safety in India.

🎯 Learning Objectives

After completing this module, you will be able to:

- Understand why helmets and seat belts are effective, not just legally required
- Choose and install the right child safety seat for your child's age
- Understand how airbags work and their limitations
- Check tyre safety and know when tyres become unsafe
- Maintain a vehicle to keep it safe and roadworthy
- Park safely and legally
- Understand India's driving licence categories and requirements
- Know which vehicle documents you must carry, and how
- Understand vehicle fitness certificates and mandatory insurance

🚦 Why This Topic Matters

A vehicle is only as safe as the equipment inside it and the condition it is kept in. Even the most careful rider or driver remains at real risk if a helmet is uncertified, a tyre is worn smooth, or a child is left unrestrained.

> **Reality Check:** MoRTH's "Road Accidents in India 2022" report attributes nearly **30% of road fatalities to two-wheeler riders not wearing helmets**, and a certified ISI helmet cuts the risk of fatal head injury by around **42%**. Meanwhile, a 2021 survey by the SaveLIFE Foundation found that **over 75% of Indian parents** had never used a car seat for their child.

This is why understanding safety equipment and vehicle upkeep is not paperwork — it is protection, in the most literal sense.

📖 Introduction

Two-Wheeler & Four-Wheeler Safety covers everything that stands between a road user and serious harm during a crash — personal protective equipment such as helmets, seat belts, and child restraints; vehicle safety systems such as airbags; the physical condition of the vehicle itself, especially tyres; and the legal readiness that comes from proper licensing, documentation, and insurance.

These protections work at two levels:

Layer	What It Includes
Personal Protection	Helmets, seat belts, child restraint systems — protect the body directly during impact
Vehicle & Legal Readiness	Airbags, tyres, maintenance, licences, documents, insurance — reduce risk and ensure accountability

🧠 Fundamental Concepts

Before going deeper, let us understand a few foundational ideas that the rest of this module is built upon.

Active Safety vs. Passive Safety

Vehicle safety features fall into two broad groups:

- **Active Safety** — Features that help you avoid a crash in the first place, such as well-maintained brakes and tyres
- **Passive Safety** — Features that protect you once a crash is already happening, such as helmets, seat belts, child restraints, and airbags

Both layers matter. A vehicle with excellent passive safety but worn-out brakes is still dangerous, and a well-maintained vehicle offers little protection to an unbelted occupant in a crash.

Key Definitions

- **ISI Mark** — A certification mark from the Bureau of Indian Standards confirming a product meets a specified Indian Standard
- **Passive Restraint** — Any device, such as a seat belt or child seat, that holds an occupant securely in place during a crash
- **Child Restraint System (CRS)** — A certified seat designed to protect a child according to their age, weight, and height

- **Third-Party Insurance** — Compulsory insurance covering harm caused to other people or their property, regardless of who is at fault
- **Fitness Certificate** — Official confirmation that a vehicle meets safety and emission standards for continued road use

Complete Detailed Explanation

Part A — Helmet Safety

Under Section 129 of the Motor Vehicles Act, every rider and pillion passenger on a two-wheeler must wear a protective helmet conforming to **BIS standard IS 4151**, properly fastened. Violating this attracts a fine of ₹1,000 and a three-month licence suspension under Section 194D. Head injuries account for an estimated **60–70% of two-wheeler fatalities** in India, and a genuine ISI-marked helmet reduces the risk of fatal head injury by roughly **42%** and brain injury by around **69%**.

- Only helmets carrying a legible embossed ISI mark and BIS licence number are legal — stickers alone are not proof
- Turbaned Sikh riders are exempt from the helmet requirement under a proviso to Section 129
- Under a 2022 amendment to CMVR Rule 138, children aged 9 months to 4 years riding pillion must be secured with a crash helmet and a safety harness
- From January 2026, manufacturers must supply two BIS-certified helmets — one for the rider, one for the pillion — with every new two-wheeler sold

Part B — Seat Belt Safety

Seat belts are mandatory for the driver and every passenger, front or rear, under CMVR Rule 138(3), with a ₹1,000 fine for non-compliance. Rear seat belts are just as important as front ones: crash tests cited by road safety researchers show that in a frontal collision at around 56 km/h, unbelted rear occupants can be thrown forward with enough force to injure themselves and the front-seat occupants ahead of them. The issue gained national attention after the September 2022 death of a prominent business leader who was reportedly unbelted in the rear seat, prompting the government to draft rules mandating seat belt reminder alarms for rear seats as well.

Part C — Child Safety Seats

CMVR Rule 138, as amended in 2022, requires a certified child restraint system for every child under 135 cm in height or up to 12 years of age, and prohibits seating any child under 135 cm in the front seat.

Age / Size	Requirement
0–15 months (up to ~13 kg)	Rear-facing child seat, rear seat only
15 months–4 years (up to ~18 kg)	Forward-facing seat with a five-point harness, rear seat
4–12 years or under 135 cm	Booster seat used with the vehicle's lap-shoulder belt, rear seat

Child seats sold in India must meet the **AIS-072** standard. Non-compliance carries a ₹1,000 fine under the 2019 amendment, though awareness remains low — a 2021 SaveLIFE Foundation survey found that more than 75% of Indian parents had never used one.

Part D — Airbags

An airbag is a passive safety device that inflates within milliseconds of a sensor detecting sudden deceleration, cushioning an occupant's head and chest during a collision. India's airbag requirements have expanded in stages under **AIS-145**: a driver-side airbag became mandatory from July 2019, a front-passenger airbag from January 2022, and subsequent rules require category M1 vehicles (up to 8 seats) to carry two additional side/torso airbags for the front row and two side curtain airbags covering all outboard seats — six airbags in total, rolled out in phases for manufacturers.

Important: Airbags are designed to work together with a fastened seat belt, not instead of one. An unbelted occupant can be thrown out of position before an airbag deploys fully, reducing its protective effect and increasing injury risk.

Part E — Tyre Safety

All pneumatic tyres sold in India must carry an ISI mark under a BIS Quality Control Order in effect since May 2011. Every tyre has small raised bars inside its grooves called **Tread Wear Indicators (TWI)** — once the surrounding tread wears down to the level of these bars, the tyre has reached its minimum safe depth: **0.8 mm for two- and three-wheeler tyres** and **1.6 mm for other motor vehicle tyres**, per the Automotive Tyre Manufacturers' Association (ATMA). Beyond this point, tyres are unsafe to use.

- Check tyre pressure monthly and before long trips, including the spare
- Look for uneven wear, cuts, bulges, or embedded objects regularly
- Replace tyres once tread reaches the TWI mark, regardless of the tyre's age
- Ensure tyres on the same axle match in size and type

Part F — Vehicle Maintenance

Routine maintenance keeps active safety systems working when you need them most. This includes regular servicing, checking brakes and suspension, ensuring all lights and indicators function, keeping a valid **Pollution Under Control (PUC) certificate**, and never ignoring a dashboard warning light. As covered in Module 4, clean windshields and headlights also directly improve visibility and reduce glare-related risk at night.

Part G — Safe Parking

Module 2 of this platform covers the legal rules for where you may and may not park. From a safety standpoint, always engage the handbrake fully, turn wheels toward the kerb on a slope to prevent rolling, never park in a way that blocks another vehicle's exit or a pedestrian path, and use designated parking areas whenever available rather than the road shoulder.

Part H — Driving Licence Basics

India issues several categories of licences depending on the vehicle and purpose:

Licence Type	Key Details
Learner's Licence (LL)	Valid for 6 months; must display an "L" sign and drive accompanied by a full licence holder in the same vehicle class
Permanent Driving Licence	Issued after holding an LL for at least 30 days and passing a driving test; typically valid around 20 years or until age 50
MCWOG (gearless two-wheeler, up to 50cc)	Minimum age 16 years
MCWG / LMV (geared two-wheeler, car)	Minimum age 18 years
HMV (heavy vehicles, over 7,500 kg)	Minimum age 20 years; requires a separate commercial/transport licence for buses, trucks, and taxis
International Driving Permit (IDP)	Allows an Indian licence holder to drive in select countries abroad

Part I — Vehicle Documents

Every driver should be able to produce four core documents on request: the **Registration Certificate (RC)**, a valid **Driving Licence**, a valid **insurance certificate**, and a valid **PUC certificate**. A Press Information Bureau clarification confirms that documents stored in **DigiLocker** or the **mParivahan** app are legally equivalent to physical originals under the Information Technology Act, 2000, and traffic authorities are required to accept them during checks. Driving without a valid licence attracts a fine of ₹5,000 under the 2019 amendment, sharply higher than the earlier ₹500.

Part J — Fitness & Insurance

Fitness Certificate: Private vehicles do not require one until 15 years from initial registration, after which it must be renewed every 5 years. Commercial vehicles require a fitness test every 2 years up to 8 years of age, and annually thereafter, increasingly conducted at government-approved **Automated Testing Stations (ATS)**. Under the Vehicle Scrappage Policy, 2021, private vehicles over 15 years must pass this test to remain registered; failing twice classifies a vehicle as an End-of-Life Vehicle. Driving without a valid fitness certificate can bring a fine of ₹2,000–₹5,000 for private vehicles and up to ₹10,000 for commercial vehicles.

Insurance: Section 146 of the Motor Vehicles Act makes third-party insurance compulsory for every vehicle used on a public road, with Section 147 setting minimum coverage requirements and Section 149 obliging insurers to compensate third-party victims. Section 157 requires insurance to be transferred within 14 days of a vehicle's sale, and Section 161 provides government-funded compensation — ₹2 lakh for death, ₹50,000 for serious injury — in hit-and-run cases where the offending driver cannot be traced. Driving an uninsured vehicle is punishable under Section 196, with imprisonment of up to three months and/or a fine of ₹2,000 for a first offence, rising to ₹4,000 for a repeat offence.

📝 Important Terminologies

Term	Simple Meaning
ISI Mark	BIS certification confirming a product meets a specified Indian Standard
Active Safety	Features that help avoid a crash, such as brakes and tyres
Passive Safety	Features that protect occupants once a crash occurs, such as seat belts and airbags
Child Restraint System (CRS)	A certified seat protecting a child based on age, weight, and height
TWI	Tread Wear Indicator — a marker showing a tyre has reached minimum safe tread depth
Fitness Certificate	Official confirmation a vehicle meets safety and emission standards
Third-Party Insurance	Compulsory cover for harm caused to other people or property
PUC Certificate	Confirms a vehicle's emissions are within permitted limits
DigiLocker	A government platform for storing legally valid digital documents
IDP	International Driving Permit, allowing driving abroad on an Indian licence

🧭 Step-by-Step Understanding

Installing a Child Safety Seat Correctly

1. Choose a seat matching your child's current age, weight, and height — not just their age
2. Install it in the rear seat, using ISOFIX anchors or the vehicle's seat belt as instructed
3. Check for no more than about 2.5 cm of movement at the belt path after installation
4. Adjust harness straps to sit at or below shoulder height for rear-facing seats, and at or above for forward-facing seats
5. Re-check the fit periodically as your child grows

Checking Tyre Safety Before a Long Trip

1. Check pressure in all four tyres and the spare when tyres are cold
2. Inspect tread depth against the TWI bars in the tyre grooves
3. Look for cuts, bulges, or embedded objects on the sidewall and tread
4. Confirm both tyres on each axle match in size and type
5. Replace any tyre showing wear at or beyond the TWI mark before departing

Applying for or Renewing a Driving Licence

1. Apply for a Learner's Licence via the Parivahan Sewa portal or an RTO, and pass the basic test
2. Practise driving for at least 30 days, accompanied by a full licence holder
3. Book and pass the practical driving test for a Permanent Driving Licence
4. Renew before expiry to avoid lapses that can affect insurance claims

🇮🇳 Real-Life Indian Examples

- **The Rear Seat Belt Wake-Up Call (2022):** The death of a prominent Indian business leader, reportedly unbelted in the rear seat of a car involved in an accident, prompted the government to draft rules mandating rear seat belt reminder alarms — a stark reminder that rear seating is not automatically safer.
- **Karnataka High Court on Child Helmets (2025):** Responding to a public interest litigation, the Karnataka High Court directed authorities to ensure the availability of helmets and safety harnesses for children riding pillion, and to continue awareness campaigns — highlighting a real gap between the 2022 rule and market availability.
- **Low Awareness of Child Car Seats:** A 2021 SaveLIFE Foundation survey found that more than 75% of Indian parents were unaware of child car seats or had never used one, showing that legal requirements alone do not guarantee real-world protection without awareness.
- **The Phased Six-Airbag Rollout:** India's airbag mandate expanded step by step — driver airbag (2019), front passenger airbag (2022), and additional side and curtain airbags for M1 vehicles — showing how vehicle safety standards are deliberately phased to balance safety gains with industry readiness.

💬 Practical Scenarios

Scenario 1: You buy a helmet that looks sturdy but only has a printed sticker, not an embossed ISI mark.
Do not use it — a sticker alone is not proof of certification. Check for a legible embossed ISI mark and BIS licence number before buying.

Scenario 2: A friend in the back seat says seat belts aren't necessary since there's no steering wheel in front of them.
Insist they buckle up — an unbelted rear passenger can be thrown forward with serious force in a frontal collision, endangering themselves and others.

Scenario 3: You're taking your toddler on a short five-minute drive and consider skipping the car seat "just this once."
Use the car seat anyway — most serious crashes happen close to home, and a child restraint should never depend on trip length.

Scenario 4: Before a highway trip, you notice your tyre tread is level with the wear indicator bars.
Replace the tyre before travelling — driving beyond the TWI mark significantly increases the risk of a blowout or loss of grip.

Scenario 5: You realise your driving licence expired while you were mid-trip.
Stop driving as soon as safely possible and arrange renewal — driving on an expired licence can affect insurance claims and attracts a fine.

🏛 Government Rules / Guidelines

- **BIS IS 4151:** The national standard every legal protective helmet must meet.
- **AIS-072:** The safety standard governing child restraint systems sold in India.
- **AIS-145:** The Automotive Industry Standard covering airbags, seat belt reminders, and related occupant safety features.
- **CMVR Rule 138 (as amended, 2022):** Sets out seat belt and child restraint requirements for all seating positions.
- **Vehicle Scrappage Policy, 2021:** Requires older vehicles to pass fitness tests to remain registered, encouraging retirement of unsafe, ageing vehicles.
- **Automated Testing Stations (ATS):** Government-approved centres progressively replacing manual fitness inspections with standardised, camera- and sensor-based testing.
- **Two-Helmet Mandate (from January 2026):** Requires manufacturers to supply two BIS-certified helmets with every new two-wheeler sold.

⚖ Laws & Policies

KEY LEGAL PROVISIONS

Provision	What It Covers
Section 129	Compulsory helmet use for riders and pillion passengers
CMVR Rule 138	Seat belt and child restraint system requirements
Section 146	Compulsory third-party insurance for every vehicle
Section 147	Minimum insurance coverage requirements
Section 157	Transfer of insurance within 14 days of a vehicle sale
Section 161	Government compensation in hit-and-run cases

PENALTIES (APPROXIMATE)

Violation	Penalty
No helmet (rider or pillion)	₹1,000 fine + 3-month licence suspension (Section 194D)
No seat belt (front or rear)	₹1,000 fine
No child restraint system	₹1,000 fine
Driving without a valid licence	₹5,000 fine
No valid fitness certificate	₹2,000–₹5,000 (private); up to ₹10,000 (commercial)
Driving uninsured (Section 196)	Up to 3 months imprisonment and/or ₹2,000 (first offence); ₹4,000 (repeat)

Note: Exact fine amounts may vary slightly by state and are periodically revised. Always verify current figures with your state transport department or the Parivahan portal.

📊 Important Statistics (India)

Indicator	Figure
Road fatalities linked to helmet non-use (2022)	Nearly 30% of all road deaths (MoRTH)
Fatal head injury risk reduction from ISI helmets	Approximately 42%
Head/brain injury share of two-wheeler fatalities	60–70% of two-wheeler deaths involve head injury
Parents who never used a child car seat (2021 survey)	Over 75% (SaveLIFE Foundation)
Car occupant deaths in road crashes (2020)	17,538 — about 13% of total road fatalities (NCRB)
Minimum tyre tread depth (two/three-wheelers)	0.8 mm (ATMA / BIS)
Minimum tyre tread depth (other vehicles)	1.6 mm (ATMA / BIS)

✓ Best Practices

- Wear a genuine ISI-marked helmet with the strap fastened, every ride, every time
- Buckle every seat belt in the vehicle before moving — front and rear
- Use an age-appropriate child restraint system for every young passenger
- Check tyre pressure and tread condition regularly, not just before long trips
- Keep RC, DL, insurance, and PUC either physically or on DigiLocker/mParivahan
- Renew fitness certificates and insurance before they lapse, not after
- Address dashboard warning lights promptly rather than ignoring them

✕ Common Mistakes

- Buying a helmet based on look or price alone, without checking the ISI mark
- Assuming rear seat belts are optional or unnecessary
- Holding a child on a lap instead of using a proper child restraint system
- Treating airbags as a substitute for wearing a seat belt
- Continuing to use a tyre worn down to the TWI bars
- Letting insurance or a fitness certificate lapse and driving anyway
- Assuming a driving licence never needs renewal

🔍 Myths vs Facts

Myth	Fact
"Any helmet with a sticker is legally certified"	Only a genuine embossed ISI mark with a valid BIS licence number confirms certification
"Airbags alone are enough protection in a crash"	Airbags are designed to work together with a fastened seat belt, not as a replacement for one
"A short drive doesn't need a child seat"	Most serious crashes happen on familiar, short routes close to home
"New vehicles don't need a fitness certificate"	Private vehicles are exempt only until 15 years, after which fitness testing becomes mandatory
"Third-party insurance is optional if I drive carefully"	It is a legal requirement under Section 146 regardless of driving skill or history

📌 Do's and Don'ts

Do	Don't
Check for a genuine embossed ISI mark on helmets	Don't rely on a sticker alone as proof of certification
Buckle up every passenger, front and rear	Don't assume rear seating is inherently safer
Use an age-appropriate child restraint every trip	Don't skip it for short or "familiar" journeys
Check tyre tread and pressure regularly	Don't wait for a visible problem before checking
Keep documents valid and accessible digitally	Don't let insurance or fitness certificates lapse
Renew your licence before it expires	Don't assume an expired licence is a minor issue

🛡 Safety Tips

Helmets & Seat Belts

- Replace a helmet after any significant impact, even if it looks undamaged
- Adjust seat belts so the lap strap sits low across the hips, not the stomach

Child Safety

- Re-check the child seat's fit every few months as your child grows
- Never place a rear-facing child seat in a seat with an active front airbag

Tyres & Maintenance

- Check tyre pressure when tyres are cold, for an accurate reading
- Keep a basic toolkit and a properly inflated spare tyre in the vehicle

Documents & Insurance

- Save digital copies of all documents on DigiLocker or mParivahan as a backup
- Set a calendar reminder a month before insurance or fitness renewal dates

🎒 Practical Checklist

- [ ] My helmet carries a genuine embossed ISI mark and is properly fastened
- [ ] Every occupant's seat belt is buckled before the vehicle moves
- [ ] My child uses an age-appropriate, correctly installed restraint system
- [ ] Tyre pressure and tread depth are checked regularly
- [ ] RC, DL, insurance, and PUC are valid and accessible (physical or digital)
- [ ] My vehicle's fitness certificate is valid, if applicable
- [ ] My third-party insurance is active and up to date
- [ ] My driving licence is valid and not close to expiry

🎓 Quick Revision Notes

- Helmets must meet BIS IS 4151; genuine ISI helmets cut fatal head injury risk by ~42%
- Seat belts are mandatory for every seat, front and rear, under CMVR Rule 138(3)
- Child restraint systems are mandatory for children under 135 cm or 12 years, per age-based categories
- Airbags work with seat belts, not instead of them
- Minimum tyre tread depth: 0.8 mm (two/three-wheelers), 1.6 mm (other vehicles)
- Fitness certificates: not needed until 15 years for private vehicles; every 2 years, then annually, for commercial vehicles
- Third-party insurance is compulsory under Section 146, regardless of driving experience
- DigiLocker and mParivahan documents are legally equivalent to physical originals

🎯 Key Takeaways

1. Certified helmets and seat belts are the single most effective personal protections available
2. Child restraint systems must match a child's age, weight, and height — not just be "any" seat
3. Airbags are a complement to seat belts, never a substitute
4. Tyre condition directly affects your ability to avoid a crash in the first place
5. Valid documents and insurance are not bureaucratic formalities — they determine real support after an accident
6. Consistent maintenance and timely renewals are as much a part of safety as any single safety feature

📖 Glossary

- **Active Safety** — Features that help avoid a crash, such as brakes and tyres
- **AIS** — Automotive Industry Standard, India's technical vehicle safety standards
- **Child Restraint System (CRS)** — A certified seat protecting a child by age, weight, and height
- **DigiLocker** — A government platform for storing legally valid digital documents
- **Fitness Certificate** — Official confirmation a vehicle meets safety and emission standards
- **IDP** — International Driving Permit, for driving abroad on an Indian licence
- **ISI Mark** — BIS certification confirming a product meets a specified Indian Standard
- **Passive Safety** — Features that protect occupants during a crash, such as seat belts and airbags
- **PUC Certificate** — Confirms a vehicle's emissions are within permitted limits
- **TWI** — Tread Wear Indicator, marking a tyre's minimum safe tread depth

? Frequently Asked Questions

1. What standard must a legal helmet in India meet?

BIS standard IS 4151, confirmed by a genuine embossed ISI mark and licence number.

2. Is wearing a helmet compulsory for pillion riders too?

Yes, under Section 129 of the Motor Vehicles Act, both the rider and pillion passenger must wear a certified helmet.

3. Is a rear seat belt legally required in India?

Yes, under CMVR Rule 138(3), seat belts are mandatory for all occupants, including those in the rear seat.

4. At what age can a child sit in the front seat?

Children under 135 cm in height are prohibited from the front seat, regardless of exact age.

5. What is the child restraint requirement for infants under 15 months?

A rear-facing child seat, installed in the rear seat of the vehicle.

6. Do airbags replace the need for a seat belt?

No. Airbags are designed to work together with a fastened seat belt, not as a substitute for one.

7. When did front passenger airbags become mandatory in India?

From January 2022, following an earlier mandate for driver airbags from July 2019.

8. What is the minimum legal tyre tread depth for a car?

1.6 mm, as indicated by the tyre's built-in Tread Wear Indicator bars.

9. What is the minimum legal tyre tread depth for a two-wheeler?

0.8 mm, per ATMA and BIS guidance.

10. What is a PUC certificate?

A Pollution Under Control certificate confirming a vehicle's emissions are within permitted limits.

11. When does a private vehicle first need a fitness certificate?

Not until 15 years after initial registration, after which it must be renewed every 5 years.

12. How often do commercial vehicles need a fitness test?

Every 2 years up to 8 years of age, and annually after that.

13. Is third-party insurance compulsory in India?

Yes, under Section 146 of the Motor Vehicles Act, for every vehicle used on a public road.

14. What is the penalty for driving without insurance?

Under Section 196, up to 3 months imprisonment and/or a ₹2,000 fine for a first offence, rising to ₹4,000 for a repeat offence.

15. Are DigiLocker and mParivahan documents legally valid?

Yes. A Press Information Bureau clarification confirms they are legally equivalent to physical originals under the Information Technology Act, 2000.

16. What is the minimum age for a learner's licence on a gearless two-wheeler?

16 years, for vehicles up to 50cc.

17. How long is a Learner's Licence valid?

6 months from the date of issue.

18. What happens if insurance isn't transferred after selling a vehicle?

Section 157 requires insurance to be transferred within 14 days of sale; failing to do so can create liability and coverage issues.

19. What compensation applies in hit-and-run cases?

Government-funded compensation of ₹2 lakh for death and ₹50,000 for serious injury, under Section 161.

20. What is the penalty for driving without a valid licence?

₹5,000, under the Motor Vehicles (Amendment) Act, 2019.

21. Are children required to wear helmets on two-wheelers?

Yes — children over 4 years must wear a helmet, and children aged 9 months to 4 years riding pillion must use a crash helmet with a safety harness.

22. Where can I check my vehicle's fitness certificate status?

On the official Parivahan portal (vahan.parivahan.gov.in), using your vehicle's registration number.

🖼 Suggested Infographics, Flowcharts & Illustrations

(For the JanGyan design team — placement suggestions only; no images generated here)

- **ISI helmet mark identification guide** → within "Helmet Safety"
- **Child restraint system age/weight chart** → within "Child Safety Seats"
- **Airbag deployment diagram with seat belt** → within "Airbags"
- **TWI tyre wear indicator illustration** → within "Tyre Safety"
- **Driving licence category comparison chart** → within "Driving Licence Basics"
- **Document checklist card** (RC, DL, Insurance, PUC) → within "Vehicle Documents"

🖃 Official References

- Ministry of Road Transport and Highways (MoRTH) — "Road Accidents in India" annual report (2022 edition)
- Bureau of Indian Standards (BIS) — IS 4151 (Protective Helmets for Two-Wheeler Riders)
- Motor Vehicles Act, 1988, and Motor Vehicles (Amendment) Act, 2019 — India Code portal
- Central Motor Vehicle Rules, 1989 — Rule 138 (as amended, 2022)
- Automotive Industry Standards (AIS) — AIS-072 (Child Restraint Systems), AIS-145 (Occupant Safety)
- Press Information Bureau (PIB) — clarification on DigiLocker/mParivahan document validity, and notifications on airbag and insurance regulations
- Automotive Tyre Manufacturers' Association (ATMA) — tyre safety and Tread Wear Indicator guidance

🔗 Official Government Website Links

- Ministry of Road Transport & Highways: **https://morth.nic.in**
- Parivahan Sewa (licences, RC, fitness & challans): **https://parivahan.gov.in**
- DigiLocker: **https://digilocker.gov.in**
- Bureau of Indian Standards: **https://bis.gov.in**
- Press Information Bureau: **https://pib.gov.in**

Emergency Numbers: All-in-one Police/Ambulance/Fire: **112**

🏁 Conclusion

Safety on two wheels or four is built from small, consistent choices — a properly fastened helmet, a buckled seat belt, a correctly installed child seat, a tyre replaced before it wears too thin, and documents kept valid rather than forgotten. None of these choices are complicated, but together they form the difference between walking away from a crash and not.

This is only the beginning of your journey with JanGyan toward complete vehicle and occupant safety. Check your equipment, keep your paperwork current, and remember: **the protection you put on before you move is the only protection that works after something goes wrong.**

JanGyan — Learn Without Limits
Module 5 of the Road Safety & Public Safety Category